

The Exact Constant in a Weighted Critical Sobolev Inequality

Run fa_banach_001 — lane 14

9 August 2026

Candidate full result; likely valid. For every $\alpha > 0$, the best constant asked for in Remark 3 of arXiv:2412.15490 is independent of α and equals the sharp Euclidean three-dimensional constant

$$C_{2\alpha,6} = \sqrt{3} \left(\frac{\pi}{2} \right)^{2/3}.$$

The key is an exact transformation to the ordinary Sobolev quotient on a flat angle-excess cone, followed by sharp three-dimensional Cartan–Hadamard isoperimetry.

Source question

Giang, Tri, and Tuan [1] define

$$C_{2\alpha,6} = \inf_{u \neq 0} \frac{\left(\int_{\mathbb{R}^3} (|\nabla_x u|^2 + |x|^{2\alpha} |u_y|^2) dx_1 dx_2 dy \right)^{1/2}}{\left(\int_{\mathbb{R}^3} |x|^{2\alpha} |u|^6 dx_1 dx_2 dy \right)^{1/6}},$$

where the infimum is over their space $W_0^{1,2\alpha,6}(\mathbb{R}^3)$, the completion of $C_c^\infty(\mathbb{R}^3)$ in the sum of these two norms. Remark 3, on source PDF page 10, asks for its exact value.

10

T.H.GIANG, N.M.TRI, AND D.A.TUAN

On the other hand, we have

$$u^*(x_1, x_2, y) = \phi(r).$$

Note that the function ϕ satisfies Lemma 2. Then, from (3.3), (3.4) and Lemma 2, we deduce that

$$L_{2\alpha,6} \left(\int_{\mathbb{R}_{+1}^3} |x|^{2\alpha} |u^*|^6 dx_1 dx_2 dy \right)^{1/6} \leq \left(\int_{\mathbb{R}_{+1}^3} |\nabla_G u^*|^2 dx_1 dx_2 dy \right)^{1/2}, \quad (3.7)$$

where

$$L_{2\alpha,6} = \left(\frac{2\pi}{n(\alpha)} \right)^{\frac{-1}{3}} (\alpha + 1)^{\frac{-1}{3}} D_{2,6,3} \quad (3.8)$$

with $D_{2,6,3}$ is the constant given in Lemma 2. From (3.5) - (3.8), our Sobolev-type inequality (1.3) follows. Our proof is complete. $\square$

Remark 3. *Our proof only gives a lower bound for the best constant $C_{2\alpha,6}$. The exact value of $C_{2\alpha,6}$ remains an open question.*

Figure 1: Remark 3 in the source paper, including the surrounding normalization.

Transcription. “Our proof only gives a lower bound for the best constant $C_{2\alpha,6}$. The exact value of $C_{2\alpha,6}$ remains an open question.”

Proof intuition

The exponent and weights look anisotropic in (x_1, x_2, y) , but they are exactly the Jacobian and metric coefficients of a flat cone. If $r = |x|$, stretch the polar angle and replace r by $r^{\alpha+1}/(\alpha+1)$.

Both integrals in the quotient then become the ordinary unweighted $D^{1,2} \rightarrow L^6$ integrals on

$$X_\alpha = C_{2\pi(\alpha+1)} \times \mathbb{R}.$$

The total cone angle exceeds 2π , so the singular axis has angle excess, the metric manifestation of nonpositive curvature.

Kleiner's theorem applies to smooth Cartan–Hadamard three-manifolds, whereas X_α has a singular axis. Two elementary devices bridge this gap. First, a logarithmic cutoff removes the axis with vanishing energy. Second, every compact subset of the remaining cone exterior sits isometrically in a smooth complete nonpositively curved surface of revolution times $\mathbb{R}$. Kleiner's sharp isoperimetric inequality therefore gives the Euclidean sharp Sobolev lower bound on the cone. The reverse inequality follows because arbitrarily large genuinely Euclidean balls can be placed far from the axis.

Statement

Let

$$C_{E,3} := \inf_{0 \neq w \in C_c^\infty(\mathbb{R}^3)} \frac{\|\nabla w\|_{L^2(\mathbb{R}^3)}}{\|w\|_{L^6(\mathbb{R}^3)}} = \sqrt{3} \left(\frac{\pi}{2}\right)^{2/3}.$$

Theorem 1. *For every $\alpha > 0$,*

$$C_{2\alpha,6} = C_{E,3} = \sqrt{3} \left(\frac{\pi}{2}\right)^{2/3}.$$

Equivalently, every $u \in W_0^{1,2\alpha,6}(\mathbb{R}^3)$ satisfies

$$C_{E,3} \left(\int_{\mathbb{R}^3} |x|^{2\alpha} |u|^6 \right)^{1/6} \leq \left(\int_{\mathbb{R}^3} (|\nabla_x u|^2 + |x|^{2\alpha} |u_y|^2) \right)^{1/2},$$

and no larger constant works.

Exact transformation to a flat cone

Put $a = \alpha + 1 > 1$. Let $C_{2\pi a}$ be the flat two-dimensional cone whose polar coordinates are (ρ, φ) with $\rho > 0$ and $\varphi \in \mathbb{R}/(2\pi a\mathbb{Z})$, and set

$$X_a = C_{2\pi a} \times \mathbb{R}, \quad g_a = d\rho^2 + \rho^2 d\varphi^2 + dz^2.$$

The singular axis is $A = \{\rho = 0\} \times \mathbb{R}$.

On $\mathbb{R}^3 \setminus \{x = 0\}$ use cylindrical coordinates (r, θ, y) and define

$$\rho = \frac{r^a}{a}, \quad \varphi = a\theta, \quad z = y. \tag{1}$$

Because θ is taken modulo 2π , φ is taken modulo $2\pi a$; thus (1) is a diffeomorphism onto $X_a \setminus A$. For $v(\rho, \varphi, z) = u(r, \theta, y)$,

$$d\rho = r^{\alpha} dr, \quad d\varphi = a d\theta,$$

and consequently

$$\rho d\rho d\varphi dz = r^{2\alpha+1} dr d\theta dy = |x|^{2\alpha} dx_1 dx_2 dy. \tag{2}$$

Moreover

$$u_r = r^\alpha v_\rho, \quad u_\theta = a v_\varphi, \quad u_y = v_z.$$

Using $dx_1 dx_2 dy = r dr d\theta dy$ and $r^{2a} = a^2 \rho^2$ gives

$$\begin{aligned} \int_{\mathbb{R}^3} \left(u_r^2 + \frac{u_\theta^2}{r^2} + r^{2\alpha} u_y^2 \right) r dr d\theta dy \\ = \int_{X_a} \left(v_\rho^2 + \frac{v_\varphi^2}{\rho^2} + v_z^2 \right) \rho d\rho d\varphi dz = \int_{X_a} |\nabla_{g_a} v|^2 dV_{g_a}. \end{aligned} \quad (3)$$

Equations (2) and (3) show that the source quotient is exactly the ordinary critical Sobolev quotient on X_a .

The sharp lower bound on the cone

We first record the smooth-completion device.

Lemma 1 (Smooth Cartan–Hadamard completion). *If $K \subset X_a \setminus A$ is compact, a neighborhood of K embeds isometrically into a complete simply connected smooth three-manifold of nonpositive sectional curvature.*

Proof. Write $\varphi = a\vartheta$, where ϑ is modulo 2π . Then

$$g_a = d\rho^2 + a^2 \rho^2 d\vartheta^2 + dz^2.$$

Choose a smooth nondecreasing function $\psi : [0, \infty) \rightarrow [1, a]$ which is 1 near 0 and equals a after a short transition interval $[0, s_0]$. Set

$$h(s) = \int_0^s \psi(t) dt.$$

The metric

$$ds^2 + h(s)^2 d\vartheta^2 + dz^2$$

is smooth at $s = 0$, complete, and defined on a manifold diffeomorphic to $\mathbb{R}^3$. Its only possibly nonzero sectional curvature is the curvature of the surface-of-revolution factor,

$$-\frac{h''(s)}{h(s)} = -\frac{\psi'(s)}{h(s)} \leq 0.$$

For $s \geq s_0$ one has $h(s) = as + b = a(s - c)$ for a constant $c \geq 0$. Hence this exterior, after setting $\rho = s - c$, is exactly the cone exterior with metric $d\rho^2 + a^2 \rho^2 d\vartheta^2 + dz^2$.

By shortening the transition interval, its matching cone radius $h(s_0)/a$ can be made arbitrarily small. Choose it below $\min_K \rho > 0$. The resulting exterior contains an isometric neighborhood of K . $\square$

Kleiner’s three-dimensional Cartan–Hadamard theorem [2] states that every bounded smooth domain in a complete simply connected smooth three-manifold of nonpositive sectional curvature satisfies the sharp Euclidean isoperimetric inequality. The standard coarea and Schwarz rearrangement argument then yields, for every compactly supported smooth function w on such a manifold,

$$C_{E,3} \|w\|_6 \leq \|\nabla w\|_2. \quad (4)$$

For completeness, the implication is: replace every superlevel set of $|w|$ by a Euclidean ball of equal volume; Kleiner's perimeter inequality and coarea decrease the Dirichlet integral, while the L^6 norm is preserved, and the Euclidean Aubin–Talenti inequality finishes the estimate. This sharp implication is also recorded explicitly in Muratori–Soave [3].

Let $v \in C_c^\infty(X_a \setminus A)$. Apply Lemma 1 to its support and then (4); because the embedding is isometric on a neighborhood of the support,

$$C_{E,3}\|v\|_{L^6(X_a)} \leq \|\nabla v\|_{L^2(X_a)}. \quad (5)$$

It remains to remove the restriction away from A . For $0 < \varepsilon < 1/2$, choose a radial cutoff χ_ε which is zero for $\rho \leq \varepsilon^2$, one for $\rho \geq \varepsilon$, and linear in $\log \rho$ between these radii. Thus

$$|\chi'_\varepsilon(\rho)| \leq \frac{1}{\rho \log(1/\varepsilon)} \quad (\varepsilon^2 < \rho < \varepsilon).$$

On every bounded z -interval,

$$\int |\nabla \chi_\varepsilon|^2 dV_{g_a} \leq \frac{C}{\log^2(1/\varepsilon)} \int_{\varepsilon^2}^\varepsilon \frac{d\rho}{\rho} = \frac{C}{\log(1/\varepsilon)} \rightarrow 0. \quad (6)$$

If v is bounded, smooth away from the axis, and compactly supported, then $v_\varepsilon = \chi_\varepsilon v$ satisfies

$$v_\varepsilon \rightarrow v \text{ in } L^6(X_a), \quad \nabla v_\varepsilon \rightarrow \nabla v \text{ in } L^2(X_a),$$

by (6), dominated convergence, and the product rule. The transform of every $u \in C_c^\infty(\mathbb{R}^3)$ has these properties: it is bounded at the axis and smooth off it. Passing to the limit in (5), then using (2)–(3), proves

$$C_{2\alpha,6} \geq C_{E,3}.$$

Matching upper bound and the value

The flat cone X_a contains isometric copies of Euclidean balls of arbitrarily large radius: for a prescribed radius R , place the center at cone distance much larger than R from the axis and choose an angular chart which does not meet the identified seam. Therefore every compactly supported Euclidean test function can, after a harmless dilation and translation, be inserted into X_a with its Sobolev quotient unchanged. Taking a sequence of truncated Aubin–Talenti profiles whose Euclidean quotients tend to $C_{E,3}$ gives

$$C_{2\alpha,6} \leq C_{E,3}.$$

Together with the lower bound this proves equality.

Finally, for the Euclidean extremal $U(x) = (1 + |x|^2)^{-1/2}$,

$$\int_{\mathbb{R}^3} U^6 dx = \frac{\pi^2}{4}, \quad \int_{\mathbb{R}^3} |\nabla U|^2 dx = \frac{3\pi^2}{4}.$$

Hence

$$C_{E,3} = \frac{(3\pi^2/4)^{1/2}}{(\pi^2/4)^{1/6}} = \sqrt{3} \left(\frac{\pi}{2}\right)^{2/3},$$

which completes the proof of Theorem 1. Density extends the inequality from C_c^∞ to the source space $W_0^{1,2\alpha,6}(\mathbb{R}^3)$.

Scope, novelty, and review status

The theorem determines exactly the constant asked for in Remark 3. It does not classify extremizing sequences or assert that the infimum is attained.

A bounded search through 9 August 2026 used arXiv:2412.15490, the exact paper title and open-question phrase, $C_{2\alpha,6}$, ‘weighted Sobolev best constant’, ‘Grushin sharp Sobolev constant’, ‘flat cone Sobolev constant’, and the source authors. It found the source and general sharp Cartan–Hadamard Sobolev literature, but no later exact answer or this cone identification. This is not a comprehensive priority search.

The current verdict is **likely valid**; human review is recommended. The most important checks are the exact energy identity, the smooth completion lemma, and the zero-capacity cutoff at the cone axis.

References

- [1] T. H. Giang, N. M. Tri, and D. A. Tuan, *On some Sobolev and Pólya–Szegő type inequalities with weights and applications*, arXiv:2412.15490.
- [2] B. Kleiner, *An isoperimetric comparison theorem*, Invent. Math. **108** (1992), 37–47. doi:10.1007/BF02100598.
- [3] M. Muratori and N. Soave, *Some rigidity results for Sobolev inequalities and related PDEs on Cartan–Hadamard manifolds*, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) **24** (2023), 59–103; arXiv:2103.08240.
- [4] G. Talenti, *Best constant in Sobolev inequality*, Ann. Mat. Pura Appl. (4) **110** (1976), 353–372.